THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE UNIT TEST SET B

Chapters 5–8: **Describing Motion Around Us, How Forces Affect Motion, Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. The area under a velocity–time graph gives: 1

(a) acceleration **(b)** force **(c)** displacement **(d)** velocity

2. A vehicle moving at 20 m s^{-1} is brought to rest with a uniform retardation of 5 m s^{-2}. The stopping distance is: 1

(a) 20 m **(b)** 40 m **(c)** 4 m **(d)** 80 m

3. In uniform circular motion the velocity at any point is directed: 1

(a) away from the centre **(b)** along the tangent

(c) towards the centre **(d)** along the radius

4. Newton's first law is also called the law of: 1

(a) action **(b)** inertia **(c)** gravitation **(d)** momentum change

5. Airbags in cars protect passengers by: 1

(a) making the car stop faster

(b) increasing friction

(c) reducing the car's mass

(d) increasing the time over which the passenger stops

6. The net force needed to give a body of mass 10 kg an acceleration of 3 m s^{-2} is: 1

(a) 30 N **(b)** 13 N **(c)** 60 N **(d)** 3.33 N

7. The work-energy theorem states that the work done on a body equals the change in its: 1

(a) mass **(b)** energy **(c)** momentum only **(d)** force

8. On a lever the load is 80 N at 10 cm from the fulcrum. To balance it with the effort arm 40 cm, the effort needed is: 1

(a) 40 N **(b)** 20 N **(c)** 80 N **(d)** 90 N

9. A fixed pulley is used to: 1

(a) reduce the load **(b)** increase the work done

(c) reduce the effort **(d)** change the direction of the effort

10. A source of frequency 680 Hz produces waves of wavelength 0.5 m in a medium. The speed of the wave is: 1

(a) 680.5 m s^{-1} **(b)** 340 m s^{-1} **(c)** 680 m s^{-1} **(d)** 1360 m s^{-1}

11. A sound wave of frequency 5 Hz has a time period of: 1

(a) 0.1 s **(b)** 5 s **(c)** 0.4 s **(d)** 0.2 s

12. The audible range of a human is about: 1

(a) 20 kHz to 200 kHz **(b)** 20 Hz to 20 000 Hz
(c) 200 Hz to 2000 Hz **(d)** 2 Hz to 200 Hz

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 10–15 words each.

13. An athlete completes one round of a circular track of radius 7 m in 40 s. Find the average speed ($\pi = 22/7$). 2

14. What is inertia? On what does it depend? 2

15. State the work-energy theorem. 2

16. Under what conditions is the work done by a force zero? 2

17. Define compression and rarefaction. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 30–40 words each.

18. Distinguish between the position–time graphs of uniform and non-uniform motion. 3

19. Explain with a sketch-based example how a canoe moves forward. 3

20. The velocity of a vehicle is doubled. How does its kinetic energy change? Explain. 3

21. Describe an activity to show that sound is produced by vibrations. 3

22. Explain with an example why distance travelled and displacement can be different. 3

OR

A car moves with velocity rising uniformly from 5 m s^{-1} to 15 m s^{-1} in 10 s. Find the displacement using the graph. 3

23. Using Newton's third law, explain how we are able to walk. 3

OR

A weightlifter holds a 30 kg barbell steady. Find the force she applies ($g = 9.8$ m s^{-2}). 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

24. (i) What does a velocity–time graph show for constant velocity and constant acceleration? 5
(ii) How is acceleration obtained from it?
(iii) Find the displacement for 5 m s^{-1} rising uniformly to 15 m s^{-1} in 10 s.

OR

(i) How do you plot a position–time graph? 5
(ii) What does its slope represent?
(iii) Compare graphs for uniform and non-uniform motion.

25. (i) State the work-energy theorem. 5
(ii) Derive the expression for kinetic energy.
(iii) Find the speed of a 15 000 kg aircraft stopped within 100 m by a force of 367 500 N.

OR

(i) Define power and write its SI unit. 5
(ii) Differentiate work and power.
(iii) A motor lifts 120 kg through 15 m in 20 s. Find its power ($g = 10$).

END OF QUESTION PAPER Total: 50 Marks